Lorenzo Franceschetti

Department of Mechanical Engineering | Phone: +1 (865) 404-2069
College of Engineering | Email: lfrances@umich.edu
University of Michigan | Website: https://lorfrances.github.io/

EDUCATION

University of Michigan – Ann Arbor, MI — Aug. 2023 – present
3rd year PhD candidate in Mechanical Engineering

- National Science Foundation Graduate Research Fellow (NSF GRFP) (2024)
- Advisor: Professor Massoud Kaviany

University of Tennessee – Knoxville, TN — 2019 – 2023
Bachelor of Science in Aerospace Engineering

EXPERIENCE

Graduate Researcher, Department of Mechanical Engineering
University of Michigan – Ann Arbor, MI — July 2023 – present

- Exploring innovative methods to harvest or recycle waste heat in semiconductor devices for improved device efficiency.
- Developing self-consistent electron Monte Carlo simulation to elucidate electron and phonon transport and energy conversion physics.
- Numerically investigated the thermal performance of sintered-particle and 3D-printed evaporator wicks.
- Studied the thermal hydraulics of square-channel oscillating heat pipes (OHPs).

Undergraduate Research Assistant, Department of Mechanical and Aerospace Engineering
University of Tennessee – Knoxville, TN — March 2021 – June 2023

- Created a simulation code for liquid-metal two-phase heat transfer in oscillating heat pipes (OHPs).
- Developed a computer modeling tool to estimate thermal loads on the leading edges of hypersonic vehicles.

Mutiphysics Simulation Intern, TerraPower, LLC
Seattle, WA — June 2022 – Sep. 2022

- Developed and validated a numerical (ANSYS) model for the MHD simulation of electromagnetic pump technology used in nuclear reactors.
- Formally documented findings, including research into thermal effects, in an internal company report.

PUBLICATIONS

Franceschetti, L., Shin, S., and Kaviany, M., “Phonon valleytronics: Enhanced phonon absorption by electron valley-energy filtering,” *Physical Review B*, in press.

Franceschetti, L., Kaviany, M., and Shin, S., "Heterobarrier *in situ* phonon recycling in semiconductor diodes," *Physical Review Applied*, 25, 034063, 2026.

Franceschetti, L., Kameya, Y., and Kaviany, M., "3D-printed, ceramic porous metasurface wick: Hexagonal-prism unit-cell capillary evaporator," *International Journal of Heat and Mass Transfer*, 246, 127041, 2025.

Lu, F., **Franceschetti L.**, Krippner, K., Kaviany, M., and Daimaru, T., "Analytic thermal conductance for square channel, flat plate oscillating heat pipe: CFD simulations of Taylor liquid film and experiment, *International Journal of Heat and Mass Transfer*, 241, 126711, 2025.

Franceschetti, L., Kihm, K.D., Rawal, S.P., and White, J.R., "Pulsating Heat Pipe Performance Modeling with Different Liquid Metal Coolants Under Hypersonic Aerothermal Heating," *Journal of Thermophysics and Heat Transfer*, 39 (1), 185-198, 2025.

Brickley, S., Domenech, I., **Franceschetti, L.**, Sarappo, J., and Lyne, J.E., "Investigation of Interplanetary Trajectories to Sedna," *2023 AAS/AIAA Astrodynamics Specialist Conference*, Big Sky, MT, 2023.

CONFERENCE PRESENTATIONS

International Conference on Thermoelectrics (ITS) Houston, TX
Contributed oral presentation May 2026
Topic: "Phonon valleytronics: Enhanced phonon absorption by electron valley-energy filtering" (Franceschetti, L., Shin, S., and Kaviany, M.)

International Mechanical Engineering Congress and Exhibition (ASME) Memphis, TN
Poster presentation Nov. 2025
Topic: "Heterbarrier *in-situ* phonon recycling in semiconductor diodes" (Franceschetti, L., Shin, S., and Kaviany, M.)

OTHER TALKS

University of Michigan Ann Arbor, MI
Guest lecturer for ECE620: Electronic and Optical Properties of Semiconductors March 2026
Topic: "Semiconductor electron transport: Monte Carlo method"

- Developed an instructive, interactive web applet demo

University of Michigan Ann Arbor, MI
Guest lecturer for ME539: Heat Transfer Physics March 2025
Topic: "Semiconductor fundamentals and applications"

AWARDS AND HONORS

Alexandar Azarkhin Scholarship (University of Michigan) March 2026

- Selective scholarship awarded to one Mechanical Engineering PhD student with research interests in applied mechanics.

Distinguished Engineering Project Achievement Award (The Engineers' Council) Jan. 2025

- Awarded in recognition of research contributions in publication: "Pulsating Heat Pipe Performance Modeling with Different Liquid Metal Coolants under Hypersonic Aerothermal Heating."

National Science Foundation Graduate Research Fellowship (NSF GRFP) Fall 2024 – present

- Awarded for research proposal: "*In-situ* phonon recycling in gallium nitride light-emitting diodes for improved thermal performance, power efficiency, and lighting intensity."

Departmental Fellowship (University of Michigan) Fall 2023 – Spring 2024

- Awarded funding for the first two semesters of the PhD program in the Mechanical Engineering department.

Cook Grand Challenge Honors Program (University of Tennessee) 2019 – 2023

Undergraduate research scholarship (University of Tennessee) Spring 2022

SKILLS

Software

- ANSYS (Mechanical, Fluent, CAD), VASP, Surface Evolver, Microsoft Office

Programming

- Python, Fortran, Julia, MATLAB, supercomputer resources (bash)

Experimental

- Cleanroom semiconductor device microfabrication (photolithography)

ACADEMIC SERVICE

Reviewer, *Journal of Applied Physics*, 2025